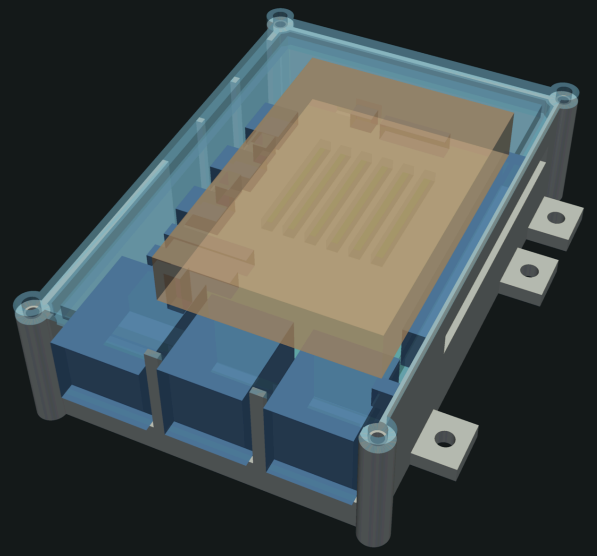
SCIENCECLAW · MIT SCIENCECLAW HACKATHON

Agents that design print-ready parts

Four specialist AI agents design a 3D-printed Raspberry Pi 5 case together, with no boss, and check it in Formlabs PreForm before it is printed.



TEAM

- Mohamed Essadak Rakah (Launch Lead): enclosure agents
- Ashutosh Mishra
- Quang Tuong Van Nguyen
- Jash Kotadiya: brackets, CalculiX simulation

THE PROBLEM

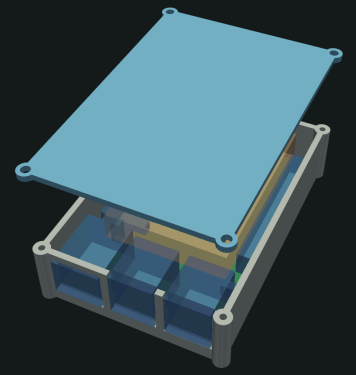
A part that looks right in CAD often doesn't fit or print

20

check failures in the naive starting design (same start for every mode)

- The board can't drop in: connectors hit the walls
- The lid sits on the USB stack; no vent over the fan
- Board screws can't grip: posts too short
- Resin-trapping cups only show up in the slicer
- Clearances depend on the printer and resin

First design (v1): naive start



The naive design (v1): closed box with port cutouts, low lid, short standoffs, no tabs, vents or cable slots.

One AI call can draw a box. Making it fit a real board, print without defects and assemble needs several kinds of expertise and real tools.

THE IDEA

Specialist agents, each owning a real tool, no boss

Design

OWNS

name, board kit

TOOL

CadQuery generator

Assembly

OWNS

box, ports, slots, tabs, lid screws

TOOL

assembly check: drop-in, lid gap, plugs, screws

Manufacturing

OWNS

walls, ribs, vents, drain holes

TOOL

printability, Formlabs PreForm, orientation tool

Fit

OWNS

clearances, standoffs, fasteners

TOOL

fit rules: design guide or measured test plate

- No boss: each agent can only change its own keys (enforced in code)
- A fix in someone else's part becomes a need to that owner; an edit is kept only if no check gets worse
- Objections must carry numbers; the protocol (plain code) decides, not a model

SYSTEM

How it works



A real exchange from the log

1 · ASSEMBLY'S CHECK (v1)

board_screw_1: no stock length gives 4.0 mm engagement within 3.5 mm of post

2 · NEED: ASSEMBLY → FIT

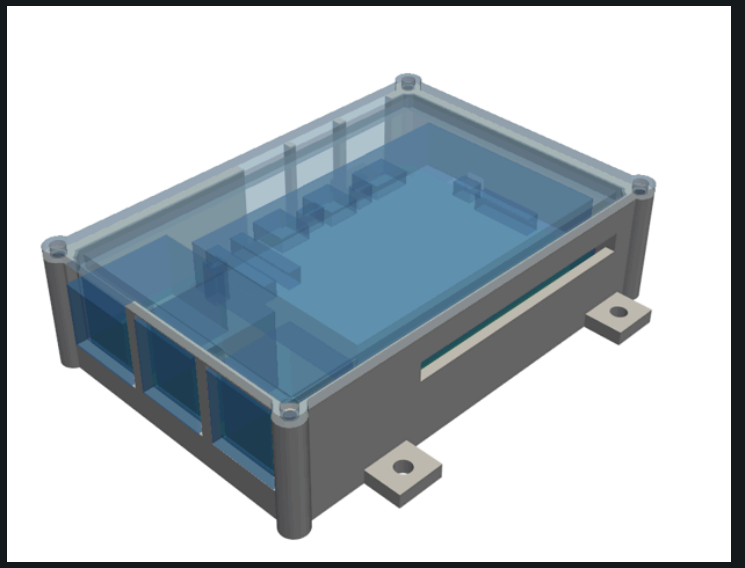
the fix is in Fit's keys, so a need is posted: suggest standoffs/height 2.9

3 · FIT'S FIX (v3)

raise standoffs 1.5 → 2.9 mm: problems 13 → 5, fixed 8, broke 0. Accepted: no check got worse

4 · COUNTER-ARGUMENT WITH NUMBERS (v5)

Fit declines a later need to lower them to 2.0 mm: it "would bring back the 0.3 mm base overlaps with microsd_socket and both cooler pin tips"

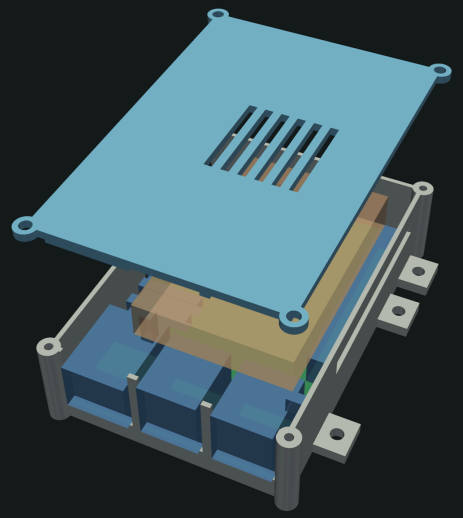


Quoted from the run log of the print candidate (Sonnet, 4 agents). This run had no objections; the decline is how Fit argued back.

RESULT · IF PRINTED

IF PRINTED: keep this slide and DELETE the next one

Print-ready result: 0 PreForm warnings



0

PreForm warnings (print job)

23.0

min print time

61.9

ml resin incl. supports

THE CUP STORY (REAL PREFORM 3.64.2)

- With PreForm's own auto-orient the final design had 1 resin-trapping cup in the base
- Manufacturing's orientation tool compared 6 orientations in PreForm; flat won for both parts
- Result: 0 warnings, both parts on one platform, 23.0 min, 61.9 ml

IF PRINTED: FILL IN

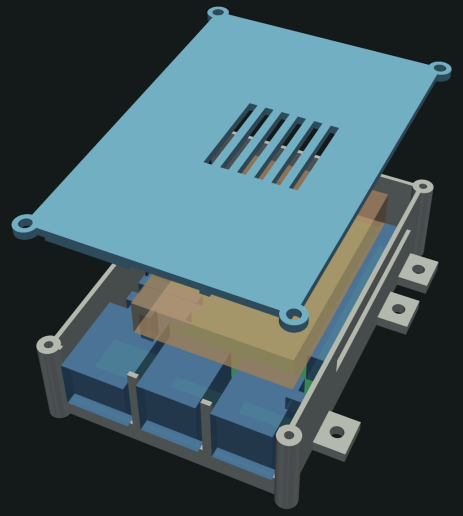
[Photo of the printed case with the Pi inside here] [Checklist from results/print/ASSEMBLY.md: board drops in __ · screws grip __/4 · lid closes __ · ports fit __/7 · fan clears __]

Say only what you tested.

RESULT · IF NOT PRINTED

IF NOT PRINTED: keep this slide and DELETE the previous one

Print-ready result: 0 PreForm warnings



0

PreForm warnings (print job)

23.0

min print time

61.9

ml resin incl. supports

THE CUP STORY (REAL PREFORM 3.64.2)

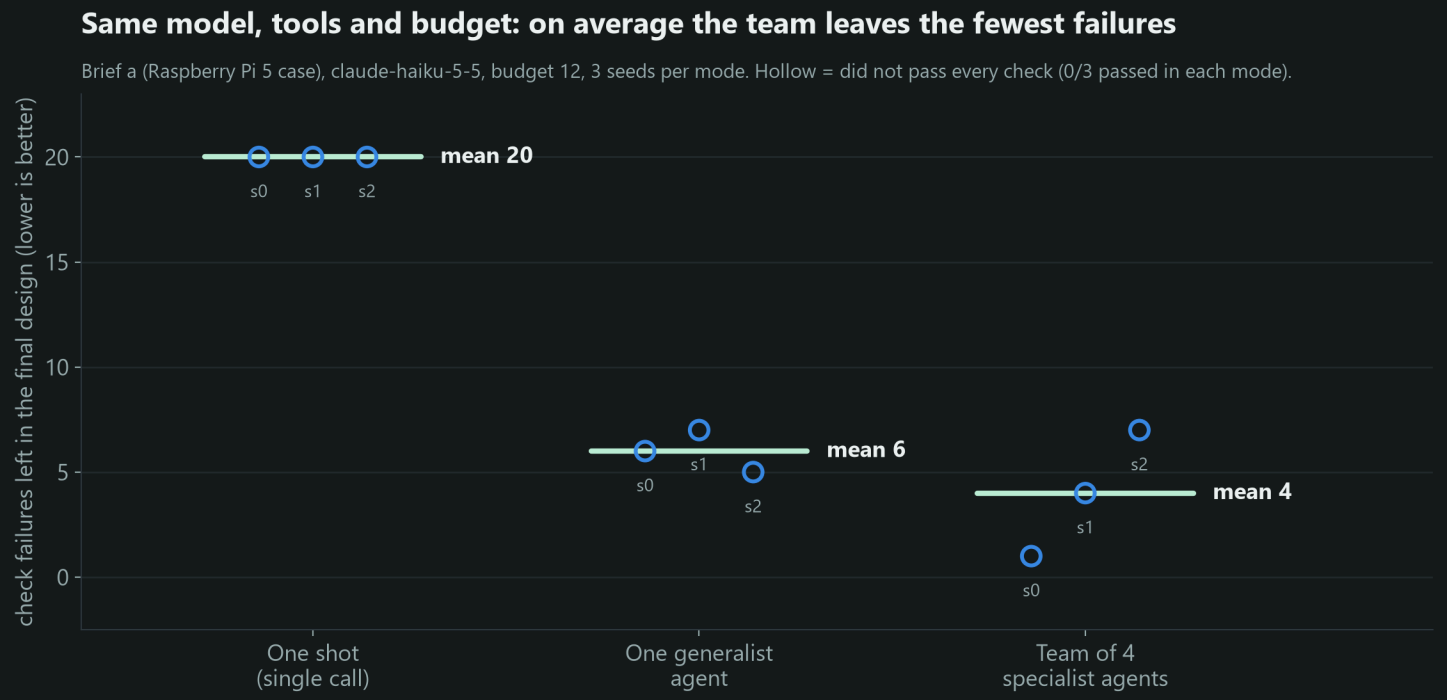
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NOT PRINTED YET: FIT CHECKED IN GEOMETRY

- Onshape: base + lid + official Pi 5 STEP in one frame; Boolean intersect = empty means no overlap
- Lid 4.4 mm over the fan keep-out, 0.8 mm over the USB stack; top risk: wide USB-A plug, upper pair, 0.07 mm (use the lower pair)

BENCHMARK · HAIKU · 3 SEEDS PER MODE

Benchmark: the team leaves the fewest failures, and no run passes



Data: results/benchmark/runs_haiku_v2_per_run.csv (from each run's summary.json / board.sqlite), runs a_pi5_standard__{one_shot,generalist,collective}__generic__s{0,1,2}__claude-haiku-5-5. Failures include the real final PreForm warnings.
Mean cost per run: one shot $0.001, generalist $0.019, team $0.032. The Sonnet benchmark is incomplete (4/18 runs) and not shown.

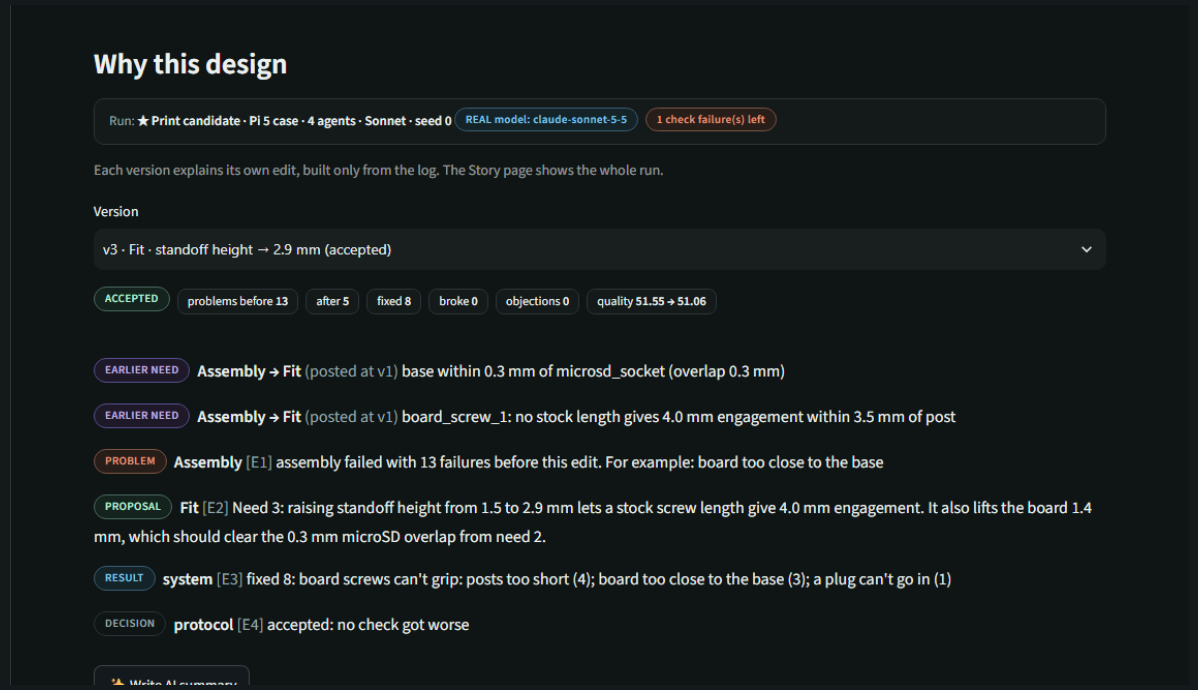
Only the Haiku set for brief a is complete. Failures include the real final PreForm warnings.

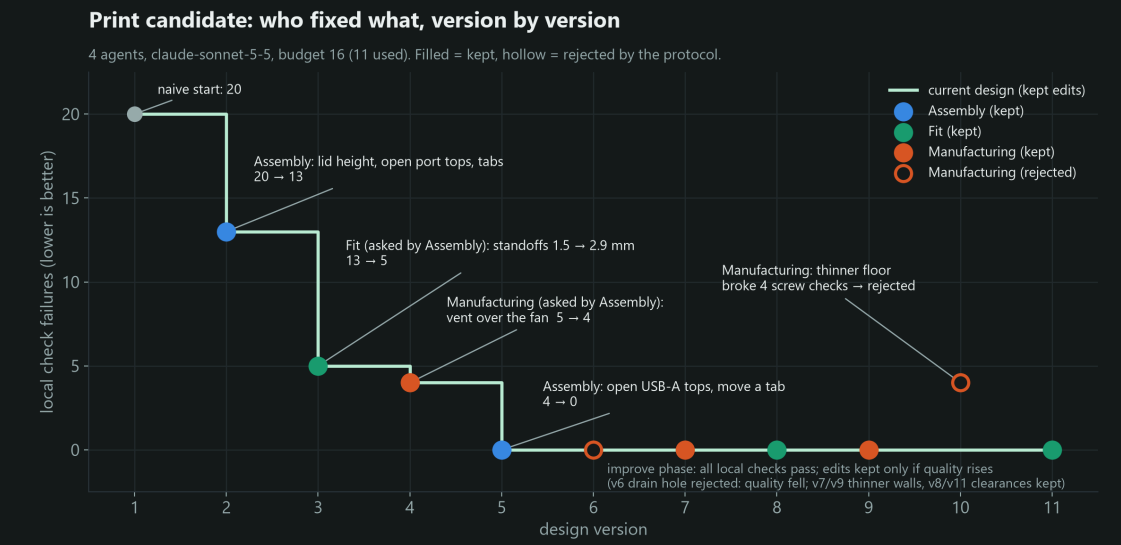
MEAN FAILURES LEFT

Team of 4: 4
One generalist: 6
One shot: 20

- Passed every check: 0/3, 0/3, 0/3
- Cost per run: team $0.032, generalist $0.019, one shot $0.001
- Same model, tools, naive start, budget 12
- Sonnet set incomplete (4/18 runs): no Sonnet claims

Trust: every claim traces back to the log





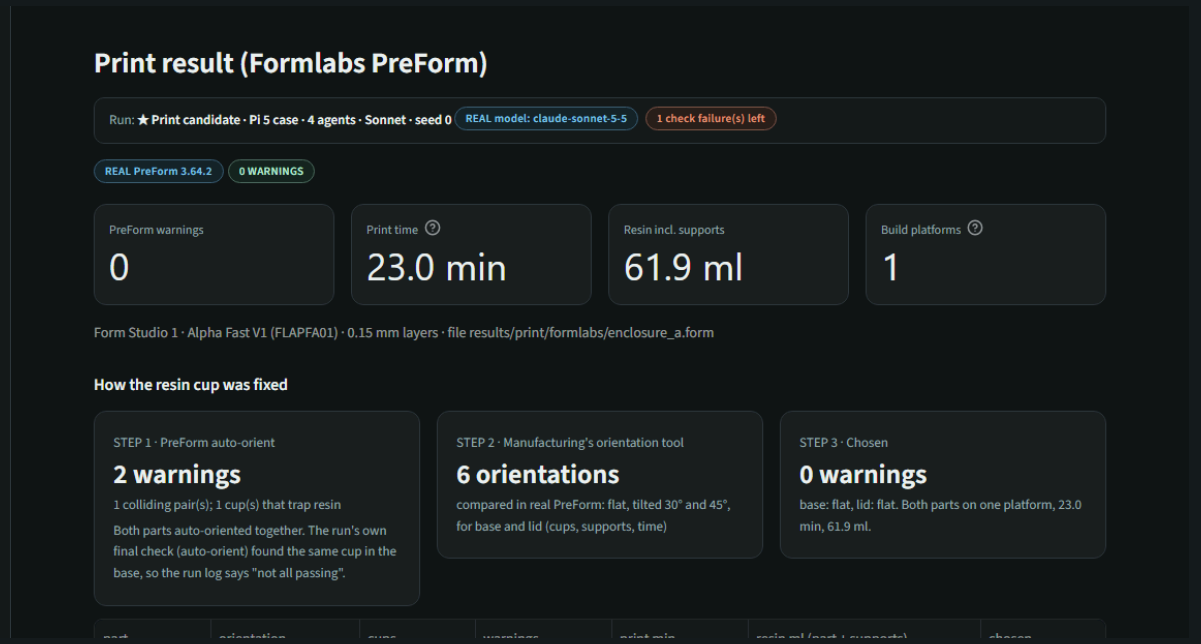
Data: results/demo_run/a_pi5_standard__collective__generic__s0__claude-sonnet-5-5/board.sqlite (assembly + printability + fit + in-loop PreForm estimate). After the run, real PreForm auto-orient still found 1 resin cup in the base;
the print job's orientation choice (Manufacturing's orientation tool) gave 0 warnings, 23.0 min, 61.9 ml (results/print/formlabs/enclosure_a.json).

- "Why this design" is built from the board log: failed check → need → proposal → breakdown → decision
- The AI summary may not write numbers: placeholders are filled from the log, any digit it writes is removed and flagged
- Charts are plain code over the logs; MOCK, UNVERIFIED and EXPERIMENTAL labels stay on everything

FORMLABS

Why it matters for Formlabs

- PreForm in the loop: the Manufacturing agent calls the PreForm Local API (PreFormServer) and acts on real warnings (cups, supports)
- Orientation tool picks the print orientation: 0 warnings, 23.0 min, 61.9 ml on one Form Studio 1 platform
- Output is a .form file that prints as is (Alpha Fast V1, 0.15 mm)
- Fit agent: clearances measured on a printed test plate for one printer + resin, not generic rules
- Same design also as SLA STL, FDM STL (not PreForm-checked) and STEP



Every PreForm number on this slide is from results/print/formlabs/enclosure_a.json (real PreForm).

TEAMMATES: REPLACE THIS SLIDE

TEAMMATES: replace this

[Brackets + CalculiX simulation]

- [Part designer: describe a bracket in words → parametric CAD → 3D finite elements (Gmsh + CalculiX)]
- [Requirements: design load, max bend, least resin; result numbers from your logs]
- [Virtual test rig vs measured bend: what was validated, what is simulated]
- [Picture: stress / bend view from the Design Lab]
- [Keep labels honest: simulated_pass vs physical_pass]

TEAMMATES: REPLACE THIS SLIDE

TEAMMATES: replace this

[Multi-provider agents]

- [Mechanics / Fabrication / Measurement specialists on different providers (OpenAI, Claude, Gemini, NVIDIA Nemotron)]
- [What a live round showed: only if it was actually run and logged]
- [Onshape / Fusion / FreeCAD link and PreForm checks on your side]
- [NVIDIA prize: only if the live Part Designer test really used Nemotron]

HONEST STATUS

Limitations and next steps

NOT YET VALIDATED

- Several sizes are UNVERIFIED (plug sizes, cooler footprint and height): ruler checks in RISK_CHECK.md
- Measured clearances need the tolerance test print (test plate not printed yet: guide values used)
- FDM version passes our checks but was not checked by PreForm
- Sonnet benchmark incomplete (4/18); no run passed every check, including real PreForm

NEXT

- Print the tolerance plate, fill in the fit test, rerun with measured clearances
- Let the orientation tool run inside the loop so runs end with 0 PreForm warnings
- Finish the benchmark (both briefs, Sonnet), compare cost per fixed failure
- More part families behind the same interface (brackets from the team, free-form EXPERIMENTAL)

TRY IT

Links

CODE (BRANCH Mo_ver: ENCLOSURE AGENTS)

https://github.com/MishraAsh2007/ScienceClaw_Project/tree/Mo_ver

DEMO VIDEO (2:50)

https://github.com/MishraAsh2007/ScienceClaw_Project/blob/Mo_ver/results/submission/demo.mp4

RUN IT WITHOUT API KEYS

pip install -r requirements.txt
streamlit run app.py (Mock mode on by default)

[Teammates: add the Jash_ver / main links here]

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4 specialist AI agents, no boss · Raspberry Pi 5 case · Formlabs PreForm: 0 warnings

